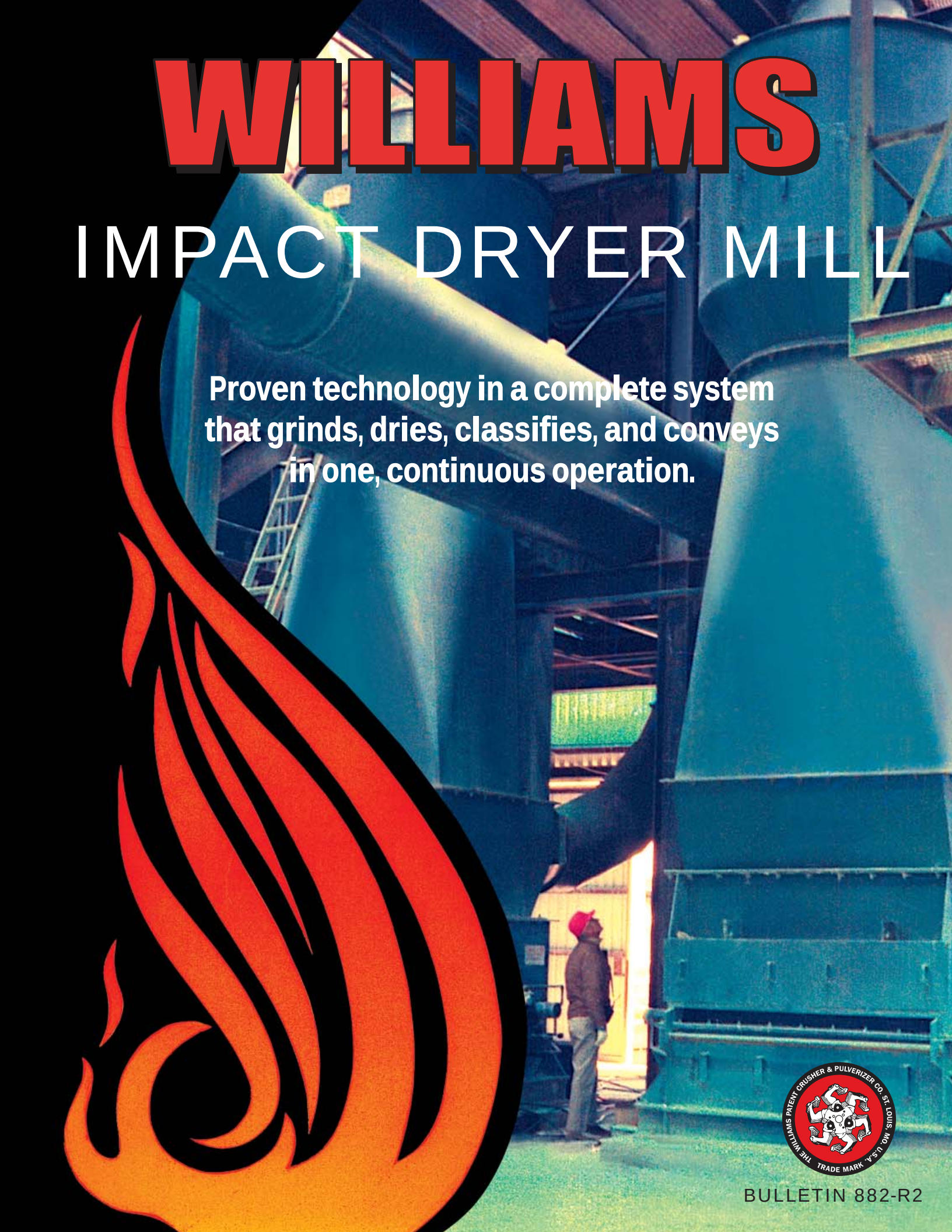
WILLIAMS

IMPACT DRYER MILL

Proven technology in a complete system that grinds, dries, classifies, and conveys in one, continuous operation.



THE WILLIAMS PATENT CRUSHER & PULVERIZER CO. ST. LOUIS, MO. U.S.A. TRADE MARK

BULLETIN 882-R2

QUICK FACTS
about
WILLIAMS
IMPACT DRYER MILLS

Lower Costs

When your process calls for both grinding and drying, a Williams Impact Dryer Mill will help you produce a premium product...at lower cost per ton. Here's how:

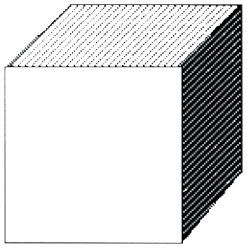
Eliminates separate drying and materials handling costs.

Reduces initial equipment investment, compared to conventional grinding methods.

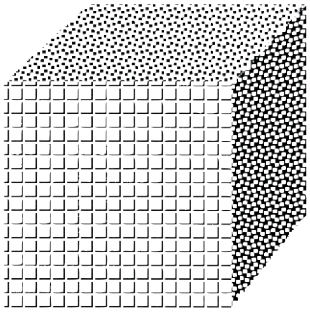
Simultaneously grinds and dries, accurately classifies, and conveys materials in one continous, automated operation, eliminating costs of separate drying and materials handling equipment.

Handles wet, sticky materials.

Dries materials faster and more efficiently, while using less fuel. Introducing heat into a Williams Impact Dryer Mill speeds drying because the material is dried in fluid bed suspension. Impact grinding the product exposes greater surface area to drying heat, the heat doesn't need to penetrate as deeply, and the entire process is completed faster. For example, pre-drying a 1-inch cube with 6 square inches of surface requires heat penetration to 1/2 inch on all sides. By comparison, a 1-inch cube divided into 200 mesh particles by Williams Impact Grinding exposed 2068 square inches, and needs heat penetration of only .00145 inch.



1'' Cube



1'' Cube Composed of 200 Mesh Particles



Improved Product

"Air Sweeping" action in the Williams Impact Dryer Mill runs a continuous stream of air through the system to "sweep" product-size particles and fines up out of the grinding chamber to the integral classifier, thus helps control grind size.

"Air Sweeping" eliminates the cushioning action of fines in the grinding chamber; conveys product size particles through the system.

Combines high output of an impact mill with accurate control of product size and moisture content without overheating.

Avoids over grinding, thus virtually eliminates excess fines in your product.

Removes impurities, using optional reject feature, to assure high-grade finished product.

Process Versatility

Williams Impact Dryer Mills can produce the following products:

Dry and grind by-product gypsum from Power Plant Scrubbers for manufacture of Wallboard.

Dry and grind filter cake (as well as other finely divided agglomerates) to a fine dry powder.

Reduce clays and other minerals, as well as pigments, to extreme fineness. Remove hard impurities if present.

Grind coal to proper fineness for burning in suspension or coal gasification.

IMPACT DRYER MILL SYSTEM

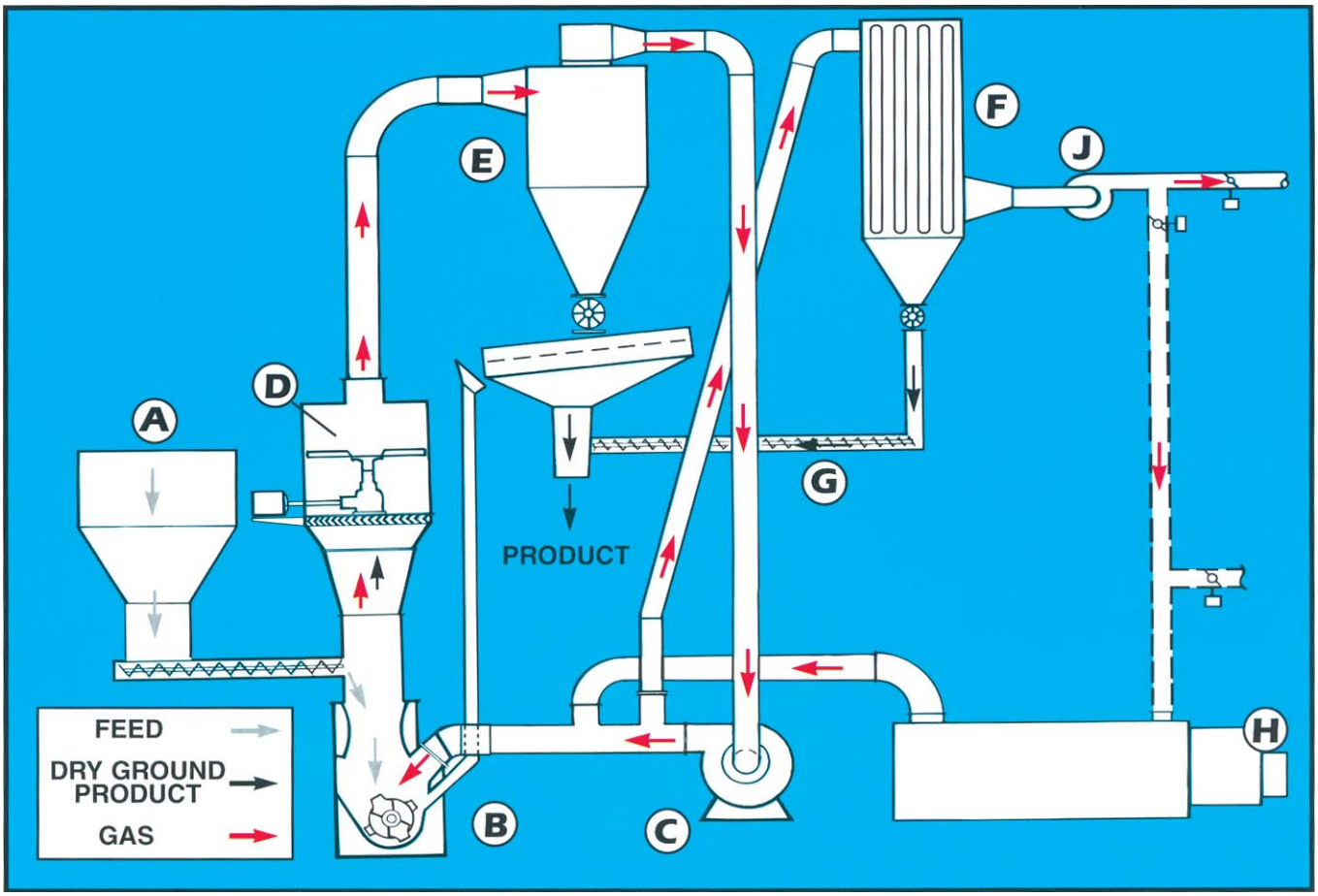
Complete automation makes grinding, drying, sizing, and conveying a "push button" operation.



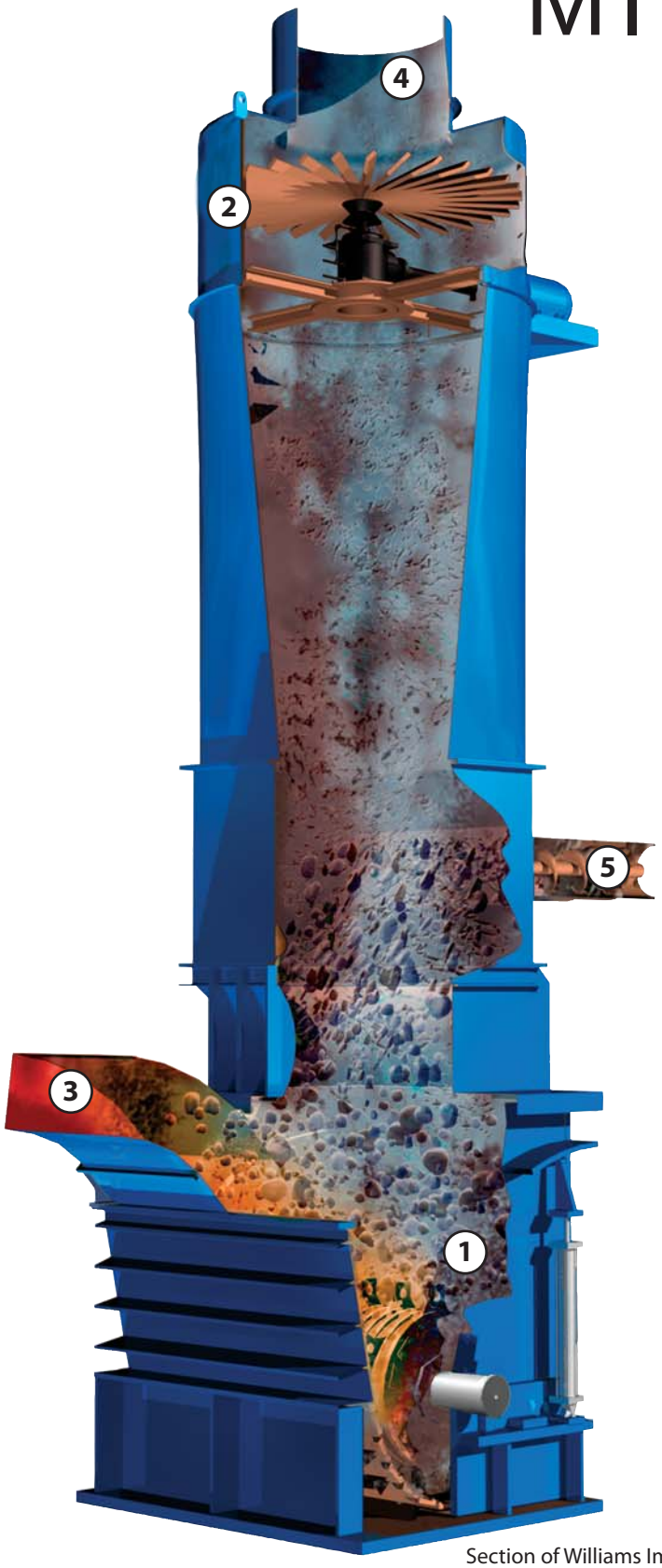
Operation of the Williams Impact Dryer Mill System is completely automated. **Feeder (A)** introduces raw material into the grinding chamber **(B)** along with heated gas at a rate that is determined by pressure and temperature sensors located in the system and adjusted automatically. In the grinding chamber, controlled action by the rotating hammers reduces feed to the desired product size. Simultaneously, the hot gas removes moisture. Drying is greatly accelerated by the increased surface area as the particle size is reduced. Controlled velocity in the grinding chamber enables the flow of hot gas to selectively air convey particles, ground to the proper size, up and out of the grinding chamber. Airflow is maintained by the **main mill fan (C).** The ground product, suspended in air, passes through the adjustable Williams Spinner Separator or **Velocity Separator (D)** where an accurate size classification is made.

Product-sized material is removed from the air-stream by the **cyclone collector (E).** Airborne fines which remain entrained in the exhaust gas from the cyclone collector are removed by a **high-efficiency fabric dust collector (F).** Extreme fines are transported by a **second conveyor (G).** An exhaust fan discharges clean gases and evaporated moisture from the system.

Patented Inert Gas System. To process explosive material such as coal, and to process all materials with greater thermal efficiency, a portion of the exhaust gas generated by **exhaust fan (J)** can be recycled to the **burner (H)** where the oxygen is consumed in the combustion process.



WILLIAMS IMPACT DRYER MILL SYSTEMS



Section of Williams Impact Dryer Mill arranged for Fine Grinding Operation.

The Standard System

The Standard System will produce a uniformly fine product in the following manner:

1. **Fluid Bed Drying and Grinding Chamber.** Controlled velocity in the grinding chamber combined with increased product surface area, resulting from size reduction, accelerates fluidized bed drying.

2. **Adjustable Spinner Separator.** Micrometer control of finished product size is possible with Williams Spinner Separator which controls product size accurately, yet permits in-process adjustment of product by means of externally mounted variable speed drive.

3. **High Capacity Air Conveying.** Gases used for drying convey the finished product from the grinding chamber to final storage. Elimination of mechanical conveying equipment thereby reduces material handling costs.

4. **Completely Enclosed System.** Dustless operation of pneumatic material handling assures cleanliness and reduces house-keeping expenses. A slight negative pressure is maintained to prevent leaks from the system.

5. **Automatic Variable-Speed Feeder.** Live-bottom bin feeders for wet, sticky materials, or rotary pocket feeders for granular materials such as coal, are controlled by conditions in the grinding chamber to provide proper feed rate automatically. The basic Williams rotary pocket feeder, variable rate vibratory feeder, or multiple screw feeder are all available to help Williams match the feeder to the job for optimum performance in your operation.

6. **Drive Motors** (not shown). Drive motors for the main mill fan, feeder, and Separator are standard, readily availabe TEFC motors.

To Produce A Fine Product...
...Or A Granular Product
with Minimum Fines

The Patented* Williams Venturi

The Patented* Williams Venturi System was developed for processing coal and other materials where specifications call for minimum extreme fines in the product.

Feed is normally introduced above the Venturi section (A). High-Velocity hot gas at the throat of the Venturi conveys material already at product size up and away from the grinding chamber.

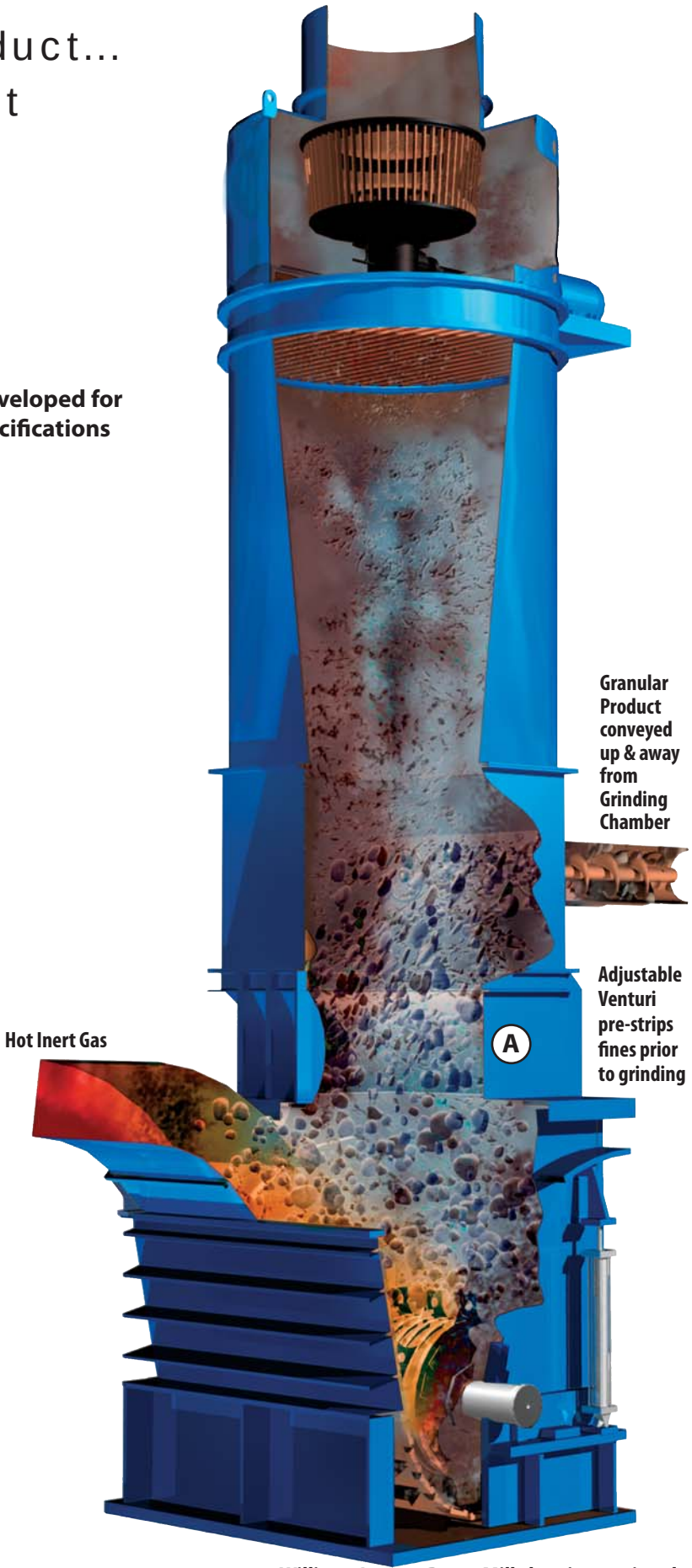
Thus, product sized particles are instantaneously removed and over reduction is prevented.

Large particles, because of their greater weight, fall through the Venturi and into the grinding chamber for reduction to specified product size.

Crushing hammers accelerate crushed particles and redirect them through the Venturi for futher particle separation.



*U.S. Patent Numbers 3,794,251 3,826,208 4,461,428 4,938,155 5,018,456 5,251,383 5,279,466 5,439,181



Williams Impact Dryer Mill showing optional Turbine Separator with Chevron Maze.

WILLIAMS
IMPACT DRYER MILL SYSTEM

OVER 125 YEARS OF PERFORMANCE AND OPERATING ECONOMY.

Free Flowing Material
Belt Feeder



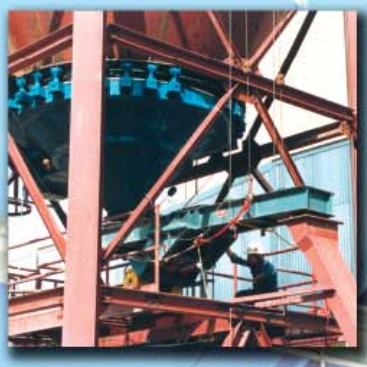
Primary Fan
700 HP Primary Fan rated at 75,000 CFM at 32" static pressure, fluidizes coal in the Impact Dryer Mill and transports product to Cyclone Collector.

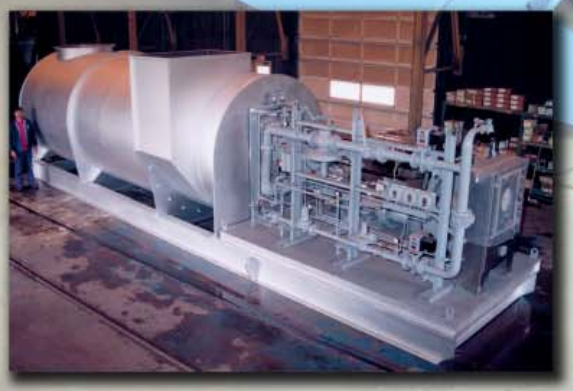


Wet Sticky Material
For feeding wet, sticky materials, a Multiple Screw Feeder forms the base of a Live Bottom Storage Bin.



Free Flowing Material
Bin Activator and Vibrating Feeder insures a free flow of material into a Williams Impact Dryer Mill.





Heater
30 million BTU per hour, gas fired heater is the source for drying and inerting gas.

Each component in our Impact Dryer Mill Systems is carefully selected to insure performance and reliability.

COMPLETE PROCESS SYSTEMS





Scalping Screen

Scalping screen insures constant product size while returning oversize material to the mill.

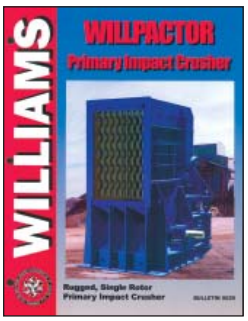


Cyclone Collectors

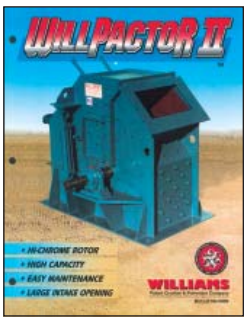
Processed coal from Impact Dryer Mill (page 2) is pneumatically conveyed by inert gas* to Cyclone Collectors and Scalping Screens.

***See patent list on page 5.**

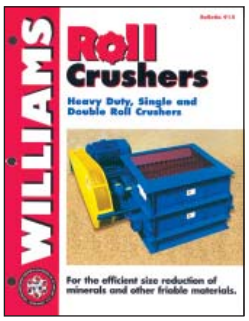
Call us to see our complete line of Williams Crushers & Equipment!



BULLETIN 862



BULLETIN 999



BULLETIN 915



BULLETIN 856



BULLETIN 998

At Williams, the company's mission is to provide highly personalized engineering that starts with a complete understanding of the size reduction need and ends with a machine designed and built to meet it.







THE WILLIAMS PATENT CRUSHER & PULVERIZER CO. ST. LOUIS, MO. U.S.A. TRADE MARK

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